

## Lab 3: Visualization

*Reading other people's figures, exploring your own data, and the one figure that answers the question*

DR. SATADISHA SAHA BHOWMICK · FALL 2026 · WEEK 04 — RELEASED MONDAY 2026-09-20, IN SECTION

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### DELIVERABLES — DUE AT THE END OF YOUR LAB SECTION, ON BRIGHTSPACE

- ☐ Lab-3\_FIRSTNAME\_LASTNAME.Rmd/Lab-3\_FIRSTNAME\_LASTNAME.qmd: the starter sheet, renamed, with the four parts completed and the generative AI disclosure filled in
- ☐ Lab-3\_FIRSTNAME\_LASTNAME.pdf: the same sheet, knitted to PDF
- ☐ the data file the sheet reads and all image files it includes

This lab is the first one run in your lab section. In this lab we focus on the figures in your analysis report. Report 1 (exploratory data analysis and visualization) is set on Tuesday 22 September and due on Friday 9 October. In this report the figures carry your argument. The lab has three parts:

- Read three figures other people made and say what each one does well or poorly.
- Explore the dataset you found in Lab 2 with different types of plots
- Choose a set of figures that most directly answer your research question. Then label them, caption them, and write a paragraph that will accompany them in your report.

about ten minutes to read, and about two hours in total

*What to download and where to put it*

Two files are on Brightspace for this lab:

- this handout;
- the starter sheet, lab-03-data-visualization.Rmd **OR** lab-03-data-visualization.qmd.

You can use any dataset you wish, but your default should be the dataset you found for Lab 2. If you have decided that your Lab 2 dataset is not deep enough to support multiple interesting figures, find another before class. Please make a folder for the lab inside the course folder (labs/lab-03/ for instance), put the .Rmd/.qmd in it, and

put the data file in a `data/` folder beside the `.Rmd/.qmd`, as in the last two labs. If you have no dataset today (Lab 2 not done, or a dataset you have since given up on), please use `PluckData` for this lab. It is on Brightspace with the week 3 and week 4 lectures. It is allowed here so you can do the lab but it is not allowed for the report.

### *What to hand in, and when*

Rename the starter to:

`Lab-3_FIRSTNAME_LASTNAME.Rmd/Lab-3_FIRSTNAME_LASTNAME.qmd`

Work through it, knit it to PDF, and submit both files on Brightspace before you leave, together with the data file the sheet reads and any images you have included in the `Rmd/qmd` file.

The `.Rmd/.qmd` and the `data/` folder have to be saved together. The sheet opens the data by a relative path and cannot find it otherwise.

#### NOTE — HOW THIS LAB IS GRADED

The syllabus rubric applies, since this lab runs in section. Labs are out of 5 points: one for arriving before the section starts, three for progress (at least half of the sheet by the end of the section), and one for the TA checking your progress when you hand it in. Half of this worksheet is Parts 1 and 2. If you miss your whole lab section in person, then submitting a fully completed worksheet/`Rmd(qmd)` file within the week means that you get three progress points and miss the two attendance points.

#### HINT

If knitting to PDF will not work: the syllabus's fallback applies: knit to HTML, convert the HTML to a PDF as shown in class, or with an online converter such as <https://www.sejda.com/html-to-pdf>, and submit that. Please also tell your TA, so that this process is sorted out before Report 1.

## Part 1: Reading a figure

Graphical displays of data are — almost always — the most effective way to communicate with a reader about a dataset. The human sensory system is immensely powerful and can process visual (or audio) information in parallel with incredible speed and fidelity. Your job as a data scientist is to select what argument you want to communicate and how to effectively support that argument with a visualization of the data. A figure that looks good and says nothing has not done the job. A plain, simple figure that makes a clear argument has totally fulfilled its job. There are a great many figures in the world that do not communicate their intended message clearly (or, indeed, have no clear intended message). Reddit has a forum where people collect these figures: [r/dataisugly](https://www.reddit.com/r/dataisugly). There is also a community that collects figures its members find attractive: [r/dataisbeautiful](https://www.reddit.com/r/dataisbeautiful). However, just because something ends up on [r/dataisbeautiful](https://www.reddit.com/r/dataisbeautiful) doesn't mean it is without reproach. You have to sharpen your eye (and your data sense) to see how a message is communicated, and whether it is done fairly and effectively. Reading figures made by other people, and critiquing them, is the quickest way to learn what to check in your own.

*What to look for*

The best way to develop your data sense is to read every figure closely. Every figure can be read with the same questions, whether it is a scatterplot from a paper or an infographic from a newspaper. Taking them in order:

1. **What is one mark?** A point, a bar, a slice, a line: what does one of them stand for? One person, one country, one year, one group? This is field 3 of the dataset card (“one row is one ...”) asked of a picture. Are there multiple marks for different types of data?
2. **What is on each axis, in what units?** A numeric axis needs a label and a unit, and it needs to be clear where zero is and whether the axis starts there. A categorical axis needs its levels named.
3. **What else is encoded?** Color, size, shape, and panels each carry a variable. Is there a legend that says which?
4. **What is the comparison?** A figure that shows one number has no comparison. Most figures are worth looking at because two things are being compared: two groups, two variables, two times.
5. **What is the claim?** In one sentence, what does the figure want you to believe? Then: is that claim visible in the marks, or only in the title?
6. **What is missing?** Sample sizes, missing values, a truncated axis, a category left out, a time period chosen to start at a low point.

Pie charts fail question 4: the reader has to compare angles, which people do badly. With more than three or four slices the comparison cannot be made at all. A bar chart of the same counts or proportions often answers the same questions with a glance. The Data to Viz caveats page collects this and the other common failures, each with an example.

A figure that easily answers those six questions is a good figure, whatever it looks like. A figure that does not have clear answers to those questions has a problem, and the question it fails is the name of the problem: “the axis has no units” or “the comparison is between angles” is a critique; “it looks confusing” is not.

**TASK 1 — THREE FIGURES**

Please find three figures and read each with the six questions. First, a bad figure from [r/dataisugly](#); second, a bad figure from [r/dataisbeautiful](#); third, a good figure from [r/dataisbeautiful](#). Three different figures, not two views of one. For each, Part 1 of the sheet asks for a link to the post (or the image itself, saved beside the sheet), then the answers to the six questions, then what is wrong or what works, and then your own attempt to say what the figure shows. Please try to read the figure even where it is hard to read: the attempt is where the problem becomes specific.

**Part 2: Exploring your own data**

Before a dataset can be explained to a reader, it has to be explored by the analyst, and those are two different kinds of figure.

Exploratory figures are for you. They are quick, plain, and numerous: every variable of interest drawn on its own, every pair of them drawn together, every comparison the question names and a few it does not. Most of them are thrown away: They are how you find out what the data hold before deciding what to say about them. An exploratory figure needs no title, no caption, and no labels beyond the column names `ggplot2` puts on the axes.

You should be generating many, many exploratory figures for your own understanding before you commit to producing anything for your report.

Explanatory figures are for the reader. There are few of them, each is chosen by you to make an argument (or one part of a larger argument). Each is labeled, captioned, and discussed. Those are the figures in the report.

### *Making excellent figures*

In the lectures, we have started drawing different kinds of figures in `ggplot2`. Every `ggplot2` has three parts: the data, a mapping from columns to the axes inside `aes()`, and at least one layer (a “geometry”) added with a plus. Here are some examples:

| To see                                                    | The call                                                                                                                                                                                                                                               |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The shape of one numeric variable                         | <code>ggplot(d, aes(x = score)) +<br/><br/>geom_histogram(bins = 20) or geom_density().<br/>Try drawing each at two bin widths (or bandwidths, using two values of <code>adjust</code>).</code>                                                        |
| A numeric variable across the levels of a categorical one | <code>ggplot(d, aes(x = group, y = score)) +<br/><br/>geom_boxplot() + geom_jitter(width = 0.15, height = 0).</code>                                                                                                                                   |
| Two numeric variables together                            | <code>ggplot(d, aes(x = hours, y = score)) +<br/><br/>geom_point(): the same three parts, with a numeric variable in <code>x</code> instead of a categorical one. Every row is one point, and the question is whether the points move together.</code> |
| Two shapes on one axis                                    | <code>ggplot(d, aes(x = score, color = group)) +<br/><br/>geom_density(): one curve per level of group in <code>ggplot2</code>.</code>                                                                                                                 |

The pictures in the Visualization section of a report are explanatory. The report is not the place for every plot you drew along the way in the same way that we don't want you to include every piece of printed output.

Two things to know about the third call: two rows with the same pair of values are drawn on the same spot, and a scatterplot of a few thousand rows is a solid mass; `geom_point(alpha = 0.3)` makes each point partly transparent, so the crowded

regions read as darker. And where one of the two variables takes only a handful of values (a rating from 1 to 5, a year), the points fall into columns. `geom_jitter()` in place of `geom_point()` spreads them, as it did for the boxplot.

#### TASK 2 — EXPLORE

Part 2 of the sheet has a chunk for each of three exploratory figures: first, the shape of each numeric variable of interest, at two widths (bin sizes or bandwidths); second, the comparison or relationship your research question names, drawn with the call from the table that fits it; third, one figure that brings in a column the question leaves out, either by putting a variable of interest against that column, or by drawing the second figure again separately for each level of a categorical column. Please draw them, and under each write one or two sentences on what you saw, and in particular on anything you did not expect: a second hump, a gap, a value that cannot be true, a group with three rows in it. Those sentences are the notes for the report's Data section and for its discussion.

#### NOTE — WHERE THE EXPLORATORY FIGURES GO

Do not put exploratory figures within the final report. You can keep them in your Rmd file, but set `include = FALSE`, or delete once you have seen them. Whatever you keep in the final report has to make a meaningful, substantial contribution to your final argument that the other figures do not make.

A categorical column that is read as text (as a character vector) works on the x axis as it is. One that was read in as a numerical code (1 and 2, say) is a number to ggplot2, and you must run `aes(x = factor(Sex))` to instruct R otherwise. The `.rds` of `PluckData` already has the labels; the `.csv` has the codes.

## Part 3: The final figures

Of the figures drawn in Part 2, a small set of figures will answer your research question more directly than the others, or they will at least come closer to answering it than any other. Those are the figures to polish and finish. They must have three things that an exploratory figure does not: axis labels in words with units, a caption, and a paragraph of text that introduces it, points at it, and says what it means.

*What one figure answers, and what a pair answers*

Every figure answers some question. A histogram of GPA answers “where do most of the grades sit, and how many students are at the ceiling?” A boxplot of phone use by sex answers “do the two groups differ in phone use?” Each of those is a real question, and each is answered by one figure. That is the test of a research question for the report. A question one figure can answer is too small for the report, because the report has three figures, and the second and third would have nothing left to add.

A question that needs two figures is a different kind of question. Take the scatterplot of GPA against phone use, which answers “do students with higher phone use have lower grades?”, and the boxplot of phone use by sex. Together they answer a question that neither answers alone: “is the drift in grades a relationship between phone use and GPA, or is it the two groups covering different ranges of phone use?” The second

figure adds something the first cannot show. It can add a variable the first figure does not have (a group, a third column), or it can add a view of the same variables that the first does not give (the shape of one variable, where the scatterplot shows only the cloud). Either way, the question needs both figures.

The report wants a question that needs three figures, each adding something the other two cannot show. Finding that question is the work of the report, not of one lab. The step we take today is the one from one figure to two.

#### TASK 3 — WHAT EACH FIGURE ANSWERS

Please look back at the three exploratory figures from Part 2 and, for each, write the question that figure alone answers: one sentence, ending in a question mark, that a reader could answer by looking at that figure and nothing else. Then choose two pairs of the figures and, for each pair, write a question that neither figure answers on its own and the two together do. Please say what the second figure adds that the first cannot show: a variable the first does not have, or a view of the same variables the first does not give. If you cannot find a pair question in your figures, that is worth knowing before Tuesday. Either the figures are too alike (draw another that brings in a column or a view the others do not have) or the research question is too small for the report and should be replaced.

#### IMPORTANT

For your final report you will have to produce three separate figures that, read together, provide a compelling argument to answer your research question. You should pick your research question (and your dataset) so that you can generate three separate figures that each build upon one another. None of the figures should say the same thing. What the three answer together, that no pair of them answers, is the question the report is written to, and that is left to you.

### *Less is more*

There is a tendency, when a report has a lot of figures, to give every pair of variables its own one. The reader then scrolls through pages of plots that share an axis and loses the thread of the argument. Where several figures share an axis, they are usually one figure: the same plot drawn once per level of a variable, side by side. In `ggplot2`, `facet_wrap(\textasciitilde Sex)` added to any of the calls above draws one panel per level of `Sex` on shared axes, and the reader compares panels by eye. The reference page has the form. A third variable can also go on color, `aes(x = hours, y = score, color = group)`, and the legend `ggplot2` adds says which level is which.

### *Labels, and a caption*

The axis labels `ggplot2` writes are the column names, which is right for an exploratory figure and wrong for the reader, who has not seen the file. `labs()` replaces them:

**R CODE**

```
ggplot(d, aes(x = hours, y = score)) +
  geom_point() +
  labs(x = "Study time (hours per week)",
       y = "Final exam score (out of 100)")
```

Each label says what the variable is, in words, and its unit in parentheses. A legend title is set the same way, with the name of the aesthetic it belongs to (color = "Section").

A caption is written as a chunk option, `fig.cap`, and when the sheet is knitted to PDF the figure is numbered and the caption is set under it:

**R CODE**

```
"\{r one-figure, fig.cap = "Final exam score against weekly
study time for 212 students, one point each. Scores rise with
study time to about ten hours a week and level off after that."\}
"
```

(The survey, the students, and the pattern in that caption are invented, as are the ones in the two captions below.) A caption is not a title. It is a description that stands on its own, so that a reader who looks only at the figure knows what is plotted, what one mark is, and what the main pattern is. Two captions for the same figure:

- Weak: "Results."
- Better: "Distribution of exam scores by study group. Students who attended the review sessions (blue) scored higher on average than students who did not (gray)."

The first tells the reader nothing. The second says what is compared, what the colors mean, and what the reader should see. It does not replace the interpretation in the text; it makes the figure readable without it.

### *The paragraph that accompanies the figure*

A figure is never left for the reader to find. The text names it before it appears ("Figure 1 shows ...", "as Figure 2 shows, the relationship between ..."), so the reader knows when to look and what to look for. Your entire argument should logically build in text, and have each of the figures introduced in making that argument. The paragraph around a figure does three things, in order:

1. **Introduce.** What the reader is about to see, and why it is here: which part of the question it addresses.
2. **Direct.** Where to look: the feature of the figure that matters, named. The second hump. The five students at 4.00. The one group whose box sits above the others.

Do use faceting (`facet_wrap()` in R or Python equivalent) for plots that share an axis. Do not draw one plot per pair of variables unless the question is about that one pair. You should have at least one faceted plot (either `facet_wrap()` or `facet_grid()` in R or Python equivalent) in your final report.

The chunk option has to stay on one logical line in the chunk header. RStudio wraps it on screen; that is fine. A line break (i.e., pressing Enter/Return to place words on a new line) typed inside the quotation marks is not.

3. **Interpret.** What that feature means for the research question, in plain language, and what it does not settle.

The caption and the paragraph do different jobs. The caption describes what the figure shows; the paragraph says what it means for the question. Please do not duplicate one in the other.

#### WARNING

Please do not present several figures in a row and interpret them together afterwards, and please do not interpret several figures in one paragraph. Each figure is followed by its own interpretation before the next one appears. A figure that has no interpretation to follow it is a figure the report does not need.

Not every figure will answer the whole question on its own. If you have a question that can be easily answered by a single figure, the question is weak and should be replaced. Instead, pick another question (or questions) with enough complexity and detail that an answer must be built towards. The pair questions you wrote above are the first step of that building: a research question the report can carry is one that a pair of figures raises and the third figure takes further.

#### TASK 4 — FINISH THE FIGURES

Part 3 of the sheet has three chunks, each with a `fig.cap` option and a paragraph under it, one per figure. Please choose the figures from Part 2 that come closest to answering your question and build on one another (the pairs you wrote questions for are the place to start), redraw each in its chunk with `labs()` (at least one of them faceted, with `facet_wrap()` or `facet_grid()`, or with a third variable on color), write the caption, and write the paragraph: introduce, direct, interpret. Please finish the first figure before starting the second; if the section ends before the third, the chunk is there for the report. Then knit, and check three things in the PDF: each figure is numbered and the caption is under it; the axes say what the variables are and in what units; and each paragraph names its figure by its number.

*When you have finished*

Knit the sheet, show the TA your work, and upload files and the data to Brightspace before you leave. What you have at the end of Part 3 is a draft of the figures in Report 1's Visualization section with captions and their paragraphs, and the notes under Part 2 are the findings that section and the Data section are built from.

#### READING — REFERENCE

- Claus Wilke, *Fundamentals of Data Visualization*, free online: [Chapter 6, Visualizing amounts](#); [Chapter 7, Visualizing distributions](#); [Chapter 9, Visualizing many distributions at once](#); [Chapter 10, Visualizing proportions](#) (and why not a pie chart); [Chapter 12, Visualizing associations among two or more quantitative variables](#); [Chapter 22, Titles, captions, and tables](#).

- [From Data to Viz](#): a decision tree from the kind of data you have to the figures that fit it, with the ggplot2 code for each, and the [caveats](#) page of common mistakes.
- The ggplot2 reference pages for [labs\(\)](#) and [facet\\_wrap\(\)](#), and the [gallery of extension packages](#) for the kinds of figure the base package does not draw.
- Lab 1, Part 2, for where the Visualization section sits in a report and what not to print; Lab 2, Parts 3 and 4, for the research question and the variables of interest the figures are built on.

### *Submission*

- ☐ Knit your document to PDF
- ☐ Check that all code is visible (not running off the page)
- ☐ Check that all output displays correctly
- ☐ Submit both .Rmd/.qmd and .pdf files